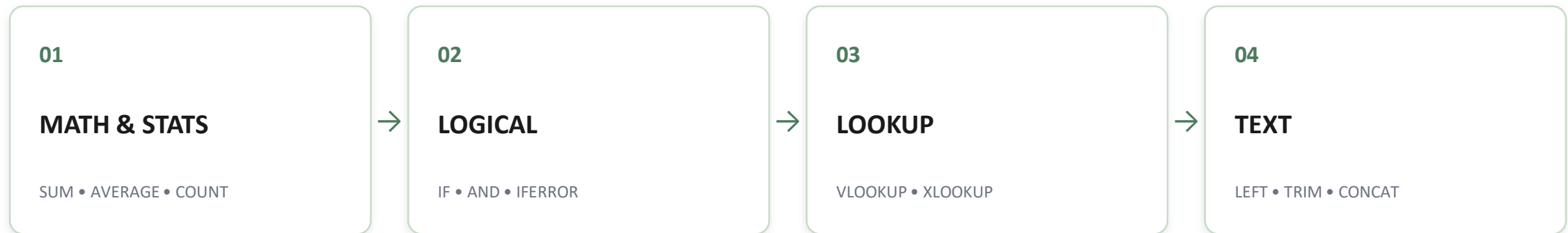


Know the **Functions** Before You Build the Formula

A simple mental map: the four families of Excel functions, what each one solves, and the exact formula to reach for.



Math & Statistical Functions

The workhorses for totals, averages, and counts across a range of cells.

 Σ

SUM

```
=SUM(range)
```

Adds every number in a range. The single most-used formula in Excel — start here for any total.

 $\bar{x}$

AVERAGE

```
=AVERAGE(range)
```

Returns the mean of a range. Ignores blank cells and text automatically.

#

COUNT / COUNTA

```
=COUNT(range) / =COUNTA(range)
```

COUNT tallies numeric cells; COUNTA tallies any non-blank cell — text included.

 $\updownarrow$

MAX / MIN

```
=MAX(range) / =MIN(range)
```

Finds the highest or lowest value in a range — quick way to spot outliers.

Logical Functions

Build decisions into a formula — test a condition, then choose what happens next.

IF

```
=IF(test, value_if_true, value_if_false)
```

The core decision function: check a condition, return one value if true, another if false.

& AND / OR

```
=AND(cond1, cond2) / =OR(cond1, cond2)
```

Combine multiple conditions — AND requires all to be true, OR requires just one.

≡ IFS

```
=IFS(test1, val1, test2, val2, ...)
```

Checks several conditions in order without nesting IF inside IF inside IF.

! IFERROR

```
=IFERROR(formula, value_if_error)
```

Catches #DIV/0!, #N/A, and other errors, replacing them with something readable.

Lookup & Reference Functions

Pull a value from one table into another by matching on a shared column.

V

VLOOKUP

```
=VLOOKUP(id, table, col_index, FALSE)
```

Searches the first column of a table for a match, then returns a value from another column. Classic, but only searches left-to-right.

X

XLOOKUP

```
=XLOOKUP(id, lookup_range, return_range)
```

The modern replacement for VLOOKUP — searches any direction and returns a clean #N/A-free result by default.

⊙

INDEX + MATCH

```
=INDEX(range, MATCH(id, id_range, 0))
```

The classic two-function combo: MATCH finds the row, INDEX returns the value — more flexible than VLOOKUP.

@

HLOOKUP

```
=HLOOKUP(id, table, row_index, FALSE)
```

Same idea as VLOOKUP, but searches across a row instead of down a column.

Text Functions

Clean, split, and combine text values before they're used elsewhere in a sheet.

+

CONCAT

```
=CONCAT (A2, " ", B2)
```

Joins text from multiple cells into one string.

LR

LEFT / RIGHT / MID

```
=LEFT (text, n)
```

Pulls a set number of characters from the start, end, or middle of text.

⌵

TRIM

```
=TRIM (text)
```

Strips extra spaces — the fix for messy, copy-pasted data.

Tt

TEXT

```
=TEXT (value, "0.0%")
```

Formats a number or date as text in a specific display format.

WHICH FUNCTION DO I ACTUALLY NEED?

Totalling numbers

SUM, AVERAGE, COUNT

Making a decision

IF, IFS, IFERROR

Pulling data from another table

XLOOKUP, VLOOKUP, INDEX + MATCH

Cleaning up text

TRIM, LEFT / RIGHT / MID, CONCAT